# Course 02323 Introduction to Statistics & 02402 Statistics (Polytechnical Foundation)

## Week 5: Hypothesis Testing

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Technical University of Denmark
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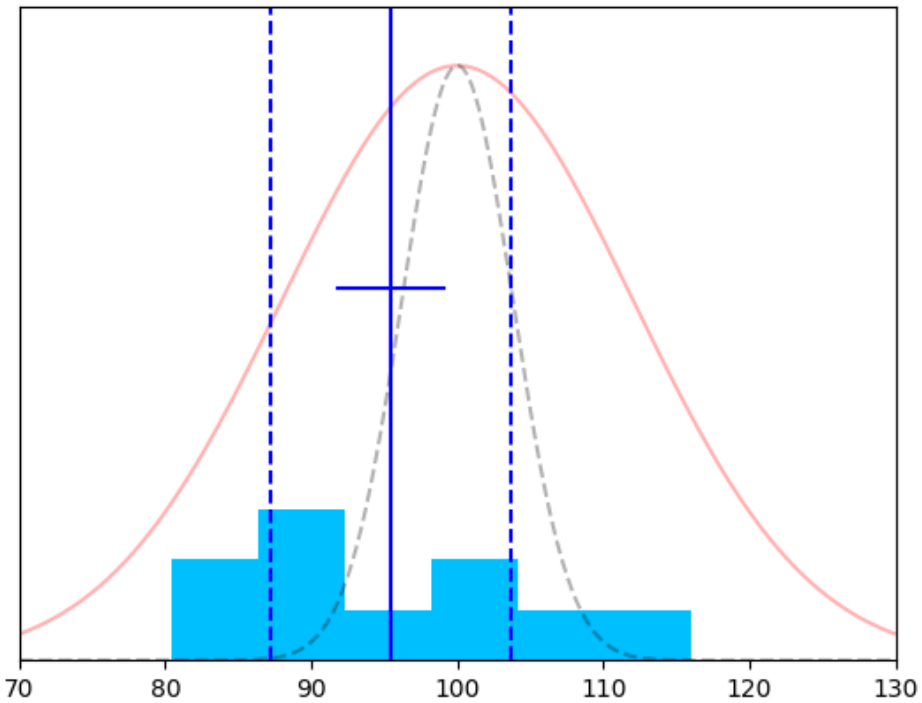
# Agenda

1. Summary from last week
2. Model Control
   - Q-Q plot
   - Transformation towards Normality
3. Hypothesis Testing
   - The Null Hypothesis and the Alternative Hypothesis
   - t-test
   - p-value
   - One-sided vs Two-sided Tests
4. Type I and Type II Errors
   - Requirements for precision
   - Power and Sample Size - One Sample

# Overview


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# From last week: Statistical inference and confidence intervals




The underlying distribution (red line) is unknown, but we can describe it with a theoretical model.

The sample mean $\bar{X}$ follows a normal distribution, $\bar{X} \sim N(\mu, \sigma/\sqrt{n})$ (black dashed line), *if* the underlying (red) distribution is itself normal OR if $n$ is large enough (Central Limit Theorem) (in that case, the red distribution can take any shape – the black will still be normally distributed).

From the sample data, we can estimate $\hat{\mu} = \bar{x}$ and calculate a corresponding confidence interval (blue dashed lines).

The confidence interval is a range within which we believe the "true" value of $\mu$ lies.

The width of the confidence interval depends on our choice of significance level ($\alpha$).

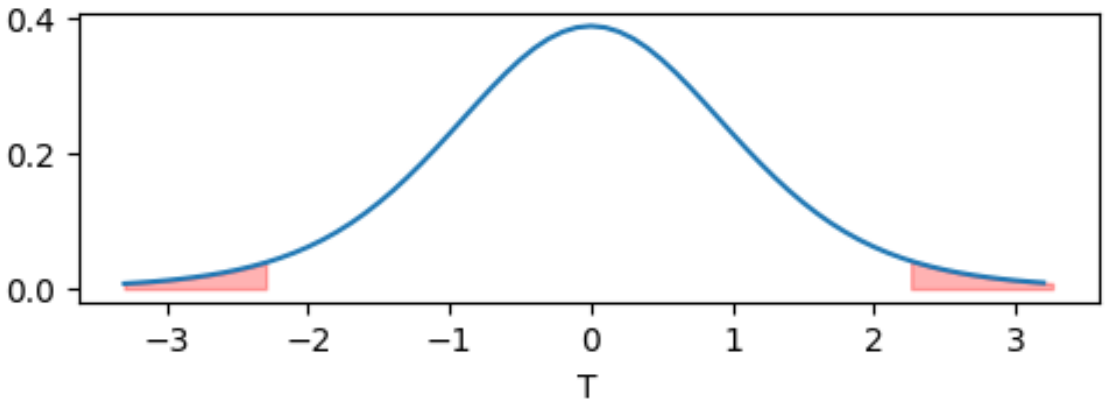
# From last week

## Confidence interval and significance level (1):

To calculate the confidence interval, we considered the statistic $T$:

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

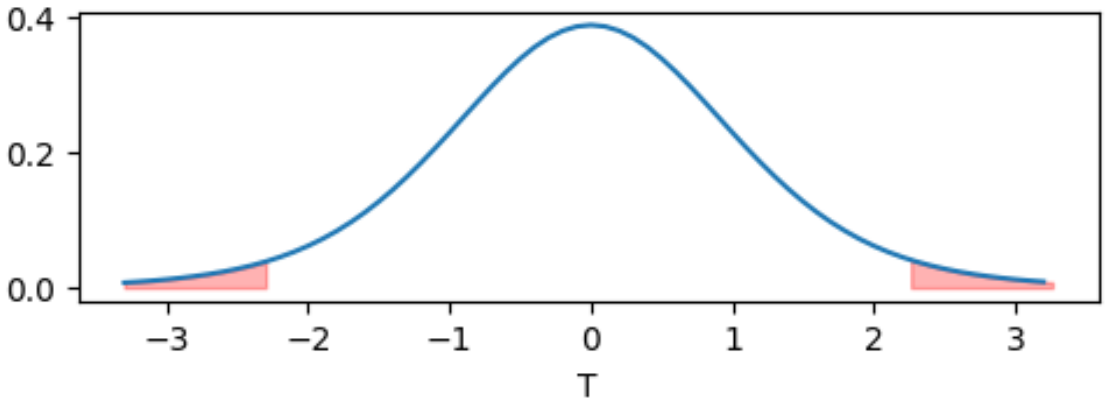
This statistic follows a t-distribution: $T \sim t(n-1)$.




It is most likely that $T$ takes values close to zero ($\bar{x}$ close to $\mu$).
"Extreme" values (large $|T|$) are less likely.

# From last week

## Confidence interval and significance level (2):




The values $\pm t_{1-\alpha/2}$ are the values of $T$ that exclude the most **extreme values** of $T$ and thus the values where $\bar{X}$ is farthest from $\mu$.

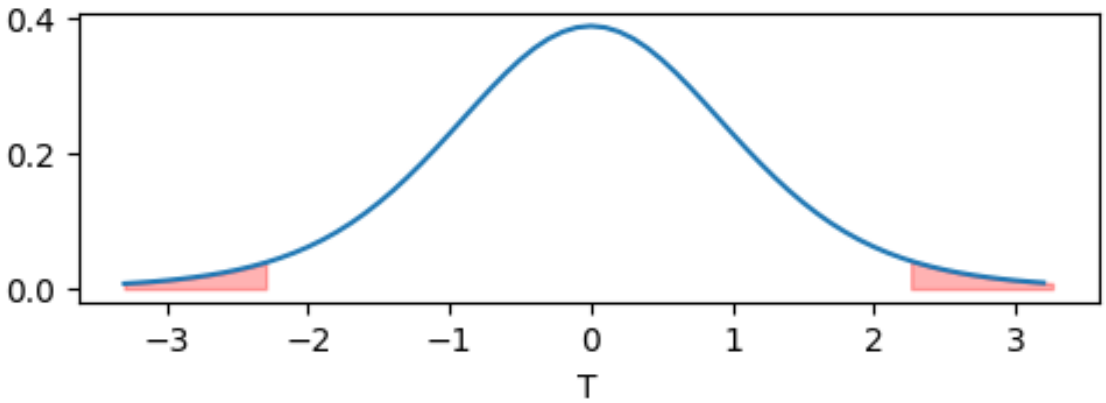
If $\alpha = 0.05$, then 95% of the *least extreme* values of $T$ fall within the boundaries $\pm t_{1-\alpha/2}$, and 5% of the *most extreme* values fall outside $\pm t_{1-\alpha/2}$ (the red region in the plot).

Kahoot!

(x2)

# From last week

## Confidence interval and significance level (3):




To calculate the confidence interval, we choose a ***significance level*** $(\alpha)$ and then calculate:

$$\left[\overline{X} - t_{1-\alpha/2}\frac{S}{\sqrt{n}}, \overline{X} + t_{1-\alpha/2}\frac{S}{\sqrt{n}}\right]$$

Larger confidence level (smaller $\alpha$) $\rightarrow$ wider confidence interval.

Kahoot!

# Overview

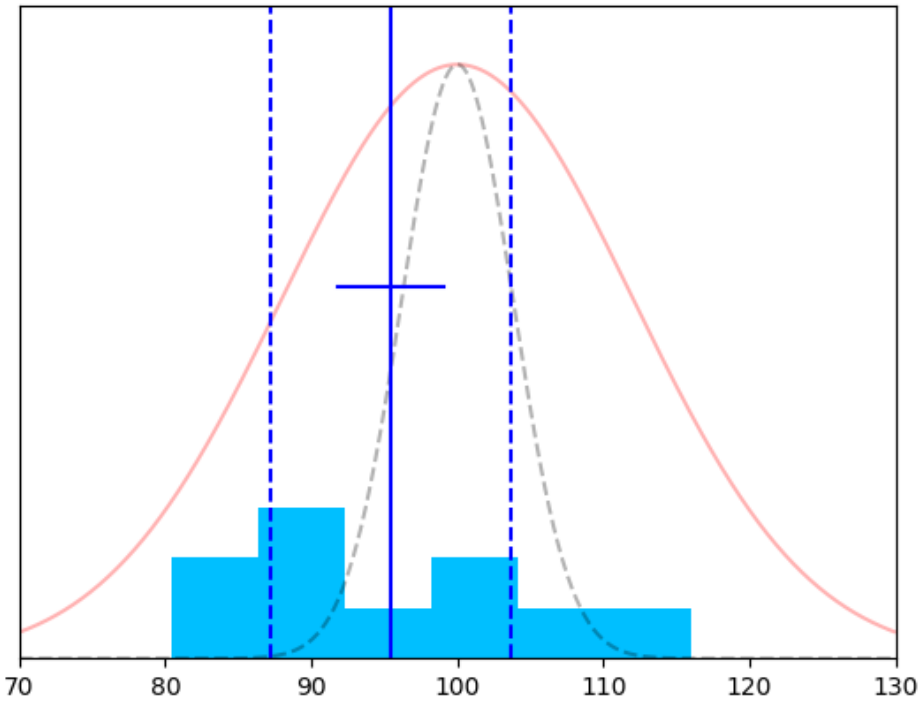

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# Checklist

After today you should be able to:

- Understand and describe the concept of choosing a significance level and performing statistical inference (including calculating confidence intervals and performing hypothesis testing).
- DATA: Perform a t-test, including calculating $t_{obs}$ and p-value, and formulate a correct conclusion.
- PYTHON: Use `stats.ttest_1samp(...)` (from the `Scipy.stats` package) to perform a t-test and calculate confidence intervals.
- THEORY: Understand the background for performing a t-test, and thereby be able to critically assess its validity (and limitations).
- Understand and use common statistical terminology, e.g. terms such as "statistical significance", "effect", "accept/reject hypothesis" and similar.
- Describe what is meant by Type 1 and Type 2 errors.
- Understand and evaluate a normal Q-Q plot.
- PYTHON: Draw a normal Q-Q plot with `sm.qqplot(...)` (from the `Statsmodels` package))
- DATA: Assess whether/not to *transform*, log-transform & median inter. est.

# Model Control




**Normal distribution assumption**:

If the sample size is small, our confidence interval calculation relies on the assumption that the underlying distribution (the red line) of the population is normally distributed. In such cases, we should verify whether this assumption holds.

We can get an idea of what the underlying distribution looks like, for instance, from a histogram of our sample data.

# Python: Sample from underlying normal distribution

- Go to today's Python notebook in VS Code
  - "Simulation: Sample from normal distribution"



# Model Control

It can be difficult to determine from a **histogram** whether the data is normally distributed—especially if the sample size is small. Additionally, the shape of a histogram depends on the choice of bin size.

Another option is to plot the **empirical cumulative distribution function** (ECDF).

You can also use a **Q-Q plot** (quantile-quantile plot). This plot compares the sorted observations $x_{(1)}, \ldots, x_{(n)}$ against the theoretical quantiles of the normal distribution.
Different definitions of quantiles exist:

- For $n > 10$, the preferred quantiles are: $p_i = \frac{i-0.5}{n}$, $i = 1, \ldots, n$
- For $n \leq 10$, the preferred quantiles are: $p_i = \frac{i-3/8}{n+1/4}$, $i = 1, \ldots, n$

# Q-Q plot

**Method 3.42 The Normal q-q plot**

The ordered observations $x_{(1)}, \ldots, x_{(n)}$, called the sample quantiles, are plotted versus a set of expected normal quantiles $z_{p_1}, \ldots, z_{p_n}$. If the points are not systematically deviating from a line, we accept the normal distribution assumption. The evaluation of this can be based on some simulations of a sample of the same size.

The usual definition of $p_1, \ldots, p_n$ to be used for finding the expected normal quantiles is

$$p_i = \frac{i - 0.5}{n}, \; i = 1, \ldots, n. \tag{3-42}$$

Hence, simply the equally distanced points between $0.5/n$ and $1 - 0.5/n$. This formula is suitable for samples with $n > 10$ and can be used in Python by specifying `qqplot(...,a=1/2)`. For samples with $n \leq 10$, the formula

$$p_i = \frac{i - 3/8}{n + 1/4}, \; i = 1, \ldots, n, \tag{3-43}$$

which can be used in Python by specifying `qqplot(...,a=3/8)`, is preferred.

# Python: ECDF and Q-Q plot

- Go to today's Python notebook in VS Code
  - "Simulation: ECDF and Q-Q plot"



# Q-Q plot

Q-Q plot is a visual/graphical inspection of data.

We have not proven that the sample comes from a normal distribution.

It can be difficult to determine whether one thinks the points in the Q-Q plot lie close enough to the straight line. To have a basis for comparison, it can therefore be a good idea to compare with simulated data (exercise).

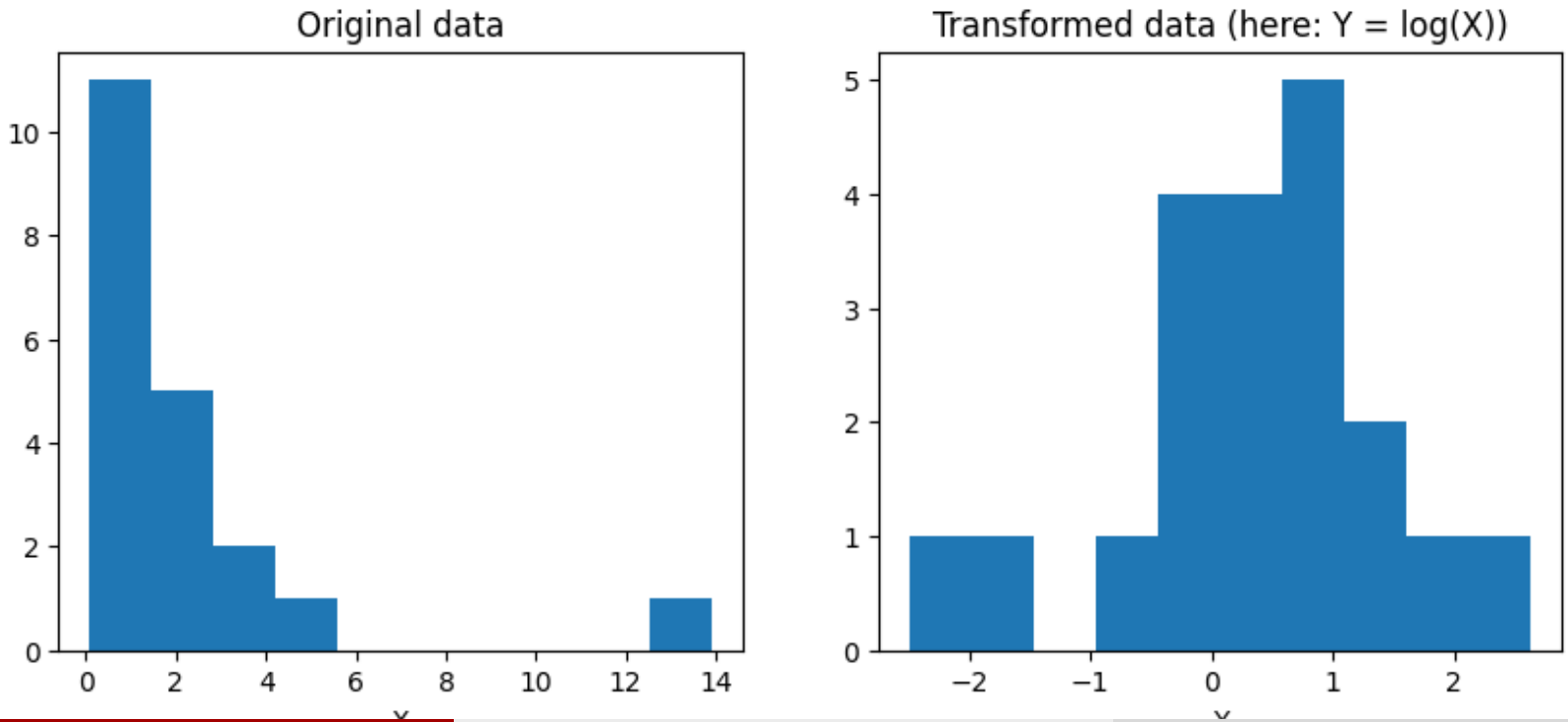
For this course's exercises on small samples, the objective is to learn the purpose of applying the method and the steps required to make the visualisation. Scientifically, the assumptions must be tested, and measures must be taken if the assumptions are not true.



# Transformation towards Normality

If data is *not* normally distributed, one option is to *transform* the data in the hope that the transformed data will better follow a normal distribution.

In such cases, calculations will be performed on the transformed data. It may be necessary to transform the results back to the original scale at the end.



# Example: Log Transformation

**Example 3.44 Radon in houses**

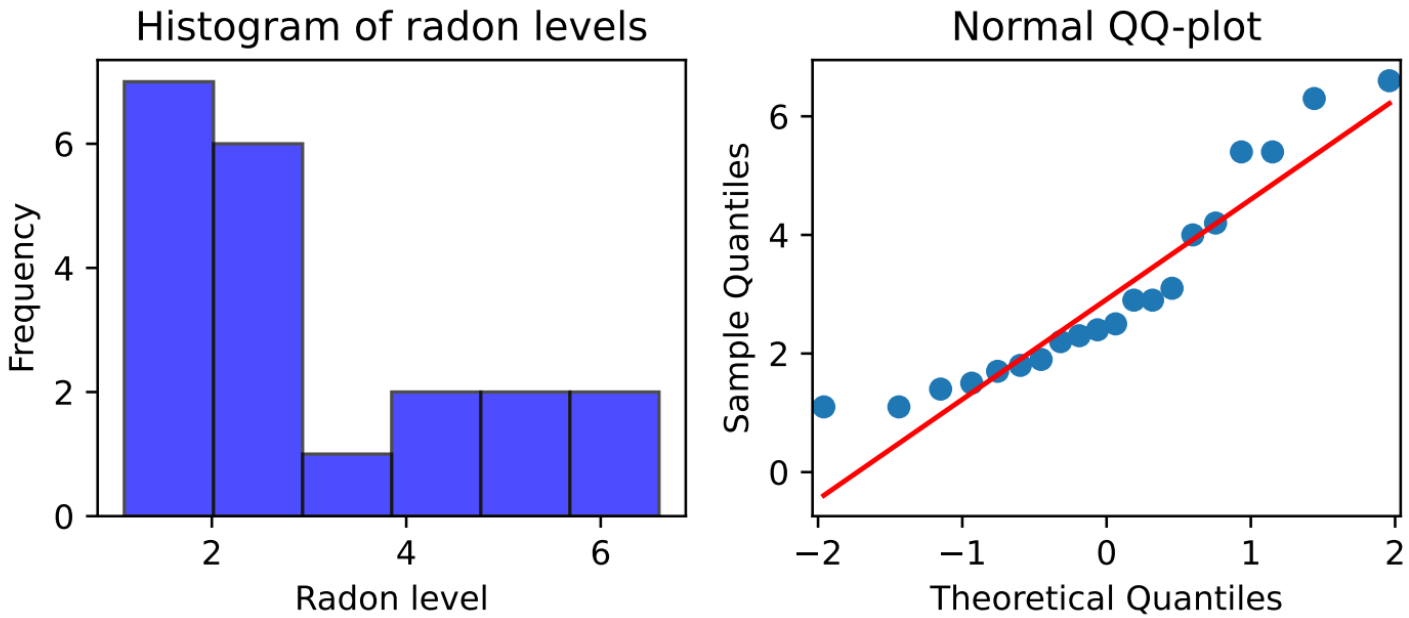
In an American study the radon level was measured in a number of houses. The Environmental Protection Agency's recommended action level is $\geq 4$ pCi/L. Here we have the results for 20 of the houses (in pCi/L):

| House | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Radon level | 2.4 | 4.2 | 1.8 | 2.5 | 5.4 | 2.2 | 4.0 | 1.1 | 1.5 | 5.4 |
| House | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| Radon level | 6.3 | 1.9 | 1.7 | 1.1 | 6.6 | 3.1 | 2.3 | 1.4 | 2.9 | 2.9 |

The sample mean, median and std. deviance is: $\bar{x} = 3.04$, $Q_2 = 2.45$ and $s_x = 1.72$.

We would like to see whether these observed radon levels could be thought of as coming from a normal distribution. To do this we will plot the data:

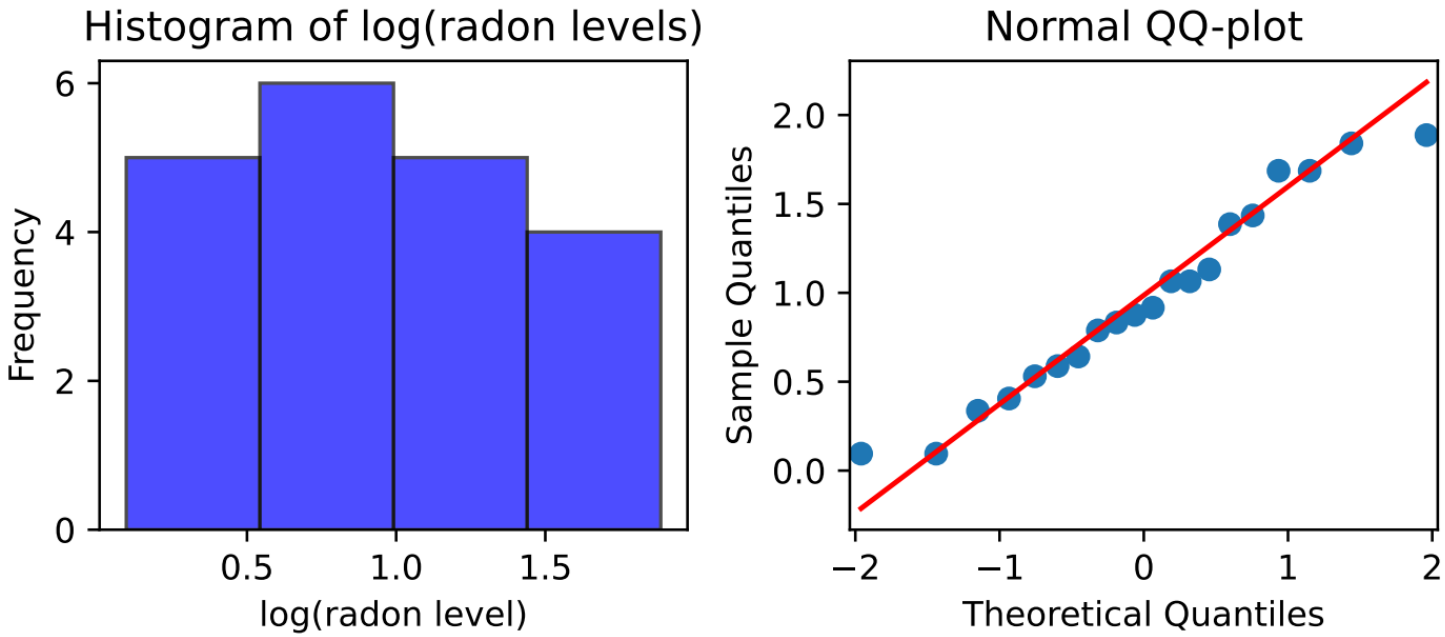
# Example: Log Transformation




From both plots we see that the data are positive and right skewed with a few large observations. Therefore a log-transformation is applied:

```
# Transform using the natural logarithm
logRadon = np.log(radon)
```

# Example: Log Transformation




As we had expected the log-transformed data seem to be closer to a normal distri-

# Example: Log Transformation

Now calculate the mean and 95% confidence interval for the log-transformed data.

Back-transform the estimated mean and confidence interval using $\exp(x)$. When we take the exponential of the estimated mean, then this is no longer a mean but a median on the original pCi/L scale. This gives a good interpretation, as *medians are useful when the distributions are not symmetric*.

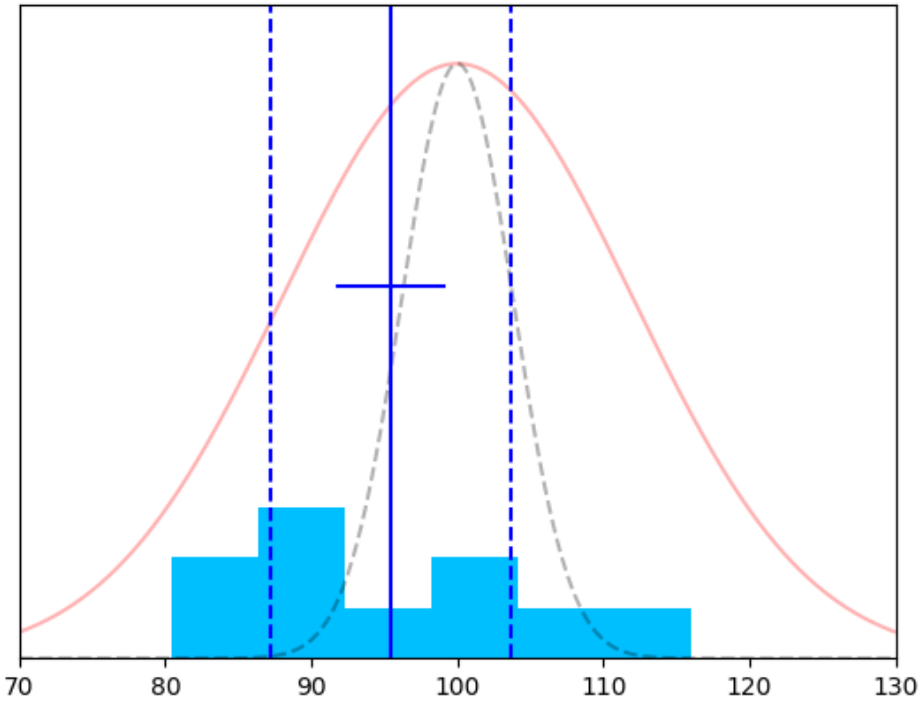
From the Python code we see that the *mean log-radon level* is 0.96 (95% CI: 0.71 to 1.22). On the original scale the estimated *median radon level* is 2.6 pCi/L (95% CI: 2.0 to 3.4).



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# Hypotheses




Let us assume that we believe the necessary assumptions hold, and we have a sample from which we estimate $\hat{\mu}$ with an accompanying confidence interval.

The plot shows a 95% confidence interval (blue dashed lines).

We do not know the underlying distribution (red) – it could be shifted more to the right or left, or it could be wider or narrower.

Today, we will discuss **hypotheses** and **tests**.

For example, one might hypothesize that $\mu = 120$. Does that seem like a reasonable hypothesis? Why or why not?

What about the hypothesis $\mu = 90$?

Which hypotheses would we "accept"? And which hypotheses would we "reject"?

# Example – Light Rail (Letbane)

## Voltage drop measurements in light rail

During the construction of a light rail, voltage drop is measured at a location where it is expected to be zero (or very small). Ten independent measurements are made, resulting in the following voltages (in volts):

Sample:

| Measurement | Voltage drop |
|---|---|
| 1 | 0.75 |
| 2 | -0.85 |
| 3 | 4.23 |
| 4 | 2.12 |
| 5 | 3.04 |
| 6 | 0.53 |
| 7 | -0.35 |
| 8 | 1.69 |
| 9 | 1.52 |
| 10 | -0.42 |

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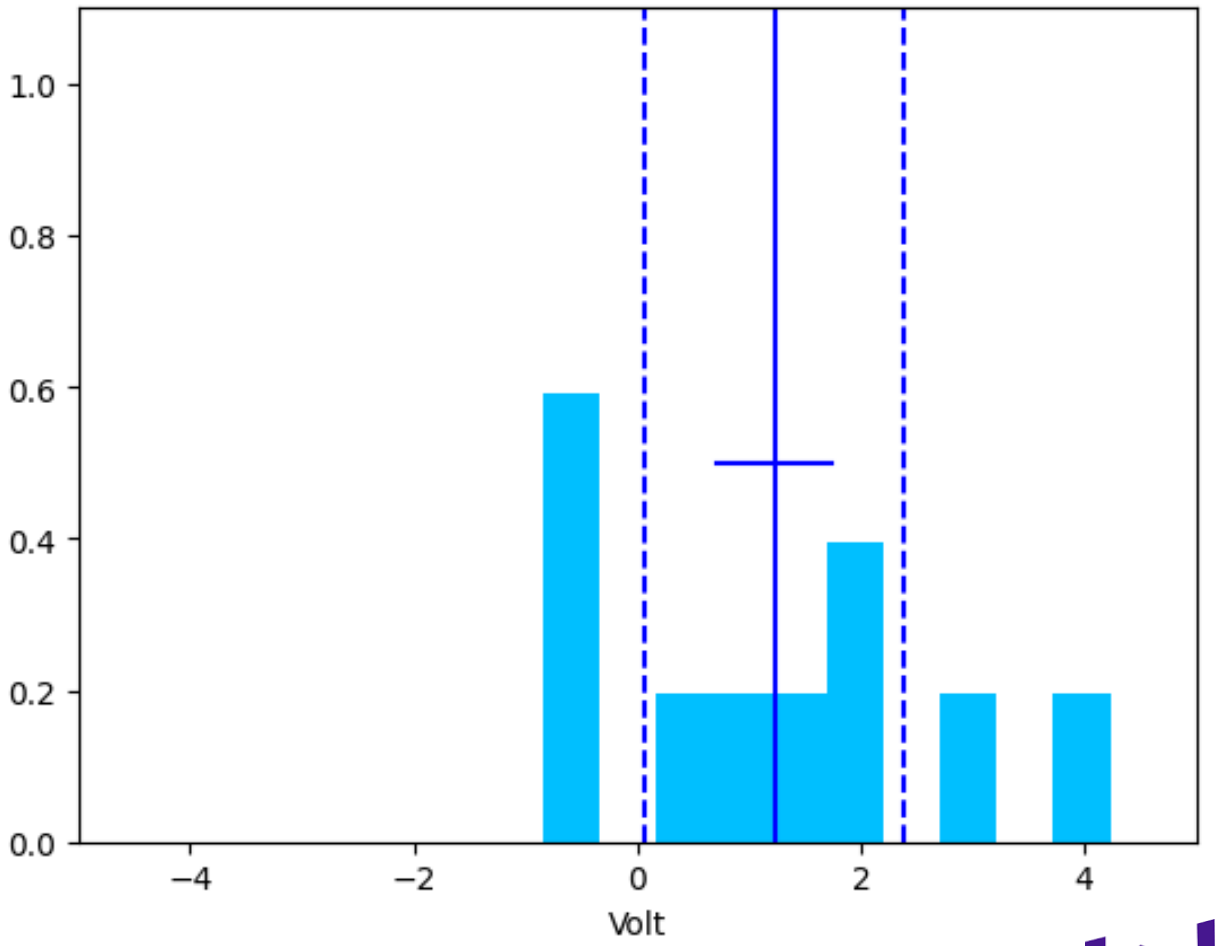
$x_i =$ voltage drop in measurement $i$

$\bar{x} = 1.23$ (sample mean)
$s = 1.62$ (sample standard deviation)

$\hat{\mu} = 1.23$

95%-confidence interval for $\hat{\mu}$:
$[0.07\,; 2.9]$

# Example - Light Rail




How could the underlying distribution look?

Could the "true" $\mu$ be 0?

Kahoot!

(x2)

# Hypothesis Test

Could the "true" $\mu$ actually be 0?

We "test" the hypothesis by calculating the *probability* of observing our sample data(*) *if* the null hypothesis were true.

(*) More precisely, we calculate the probability of a corresponding *test statistic* (a random variable).

# The Null Hypothesis and the Alternative Hypothesis

## Null Hypothesis:

We assume that $\mu$ has a specific value

$$H_0 : \ \mu = \mu_0$$

where $\mu$ is the true mean in the population.

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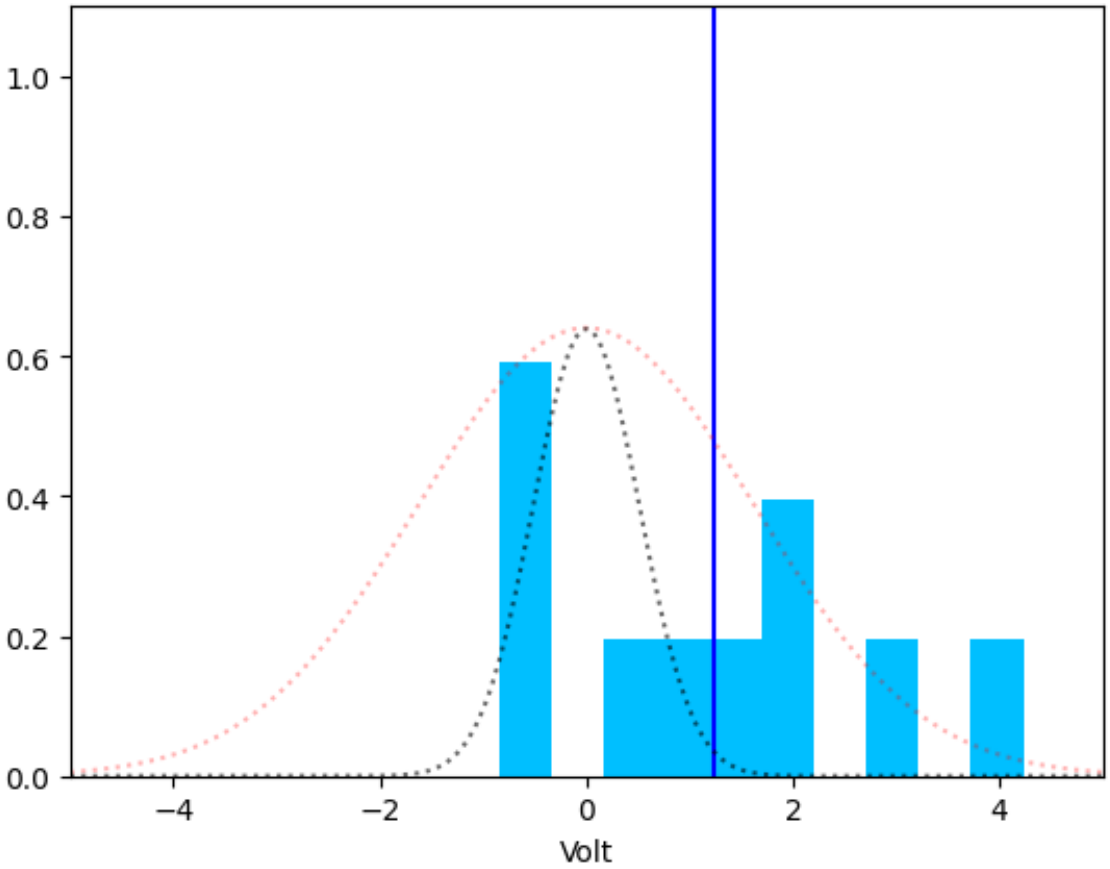
## Alternative Hypothesis:

$$H_A: \ \mu \neq \mu_0$$

## Light Rail Example:

From the data, the average voltage drop is estimated as: $\hat{\mu} = \bar{x} = 1.23$
$H_0 : \mu = 0$ (i.e., the voltage drop fluctuates around the mean zero)
$H_A : \mu \neq 0$ (i.e., the mean is not zero – and may need adjustment)

# Example - Visualization of Null Hypothesis $\mu_0 = 0$




This plot shows a normal distribution $N(0, s)$ (red) and a normal distribution $N(0, \frac{s}{\sqrt{n}})$ (black) $(s = 1.62 \text{ and } n = 10)$

# Test Statistic: $t_{obs}$

Now we will find a way to "test" our null hypothesis:

We consider the statistic $T$ under the **assumption that the null hypothesis is true**:

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$$

This statistic follows a t-distribution: $T \sim t(n-1)$.

Based on the observed data, we calculate:

$$t_{\text{obs}} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

$t_{\text{obs}}$ is our observation of the random variable $T$ (under the assumption that the null hypothesis is true).

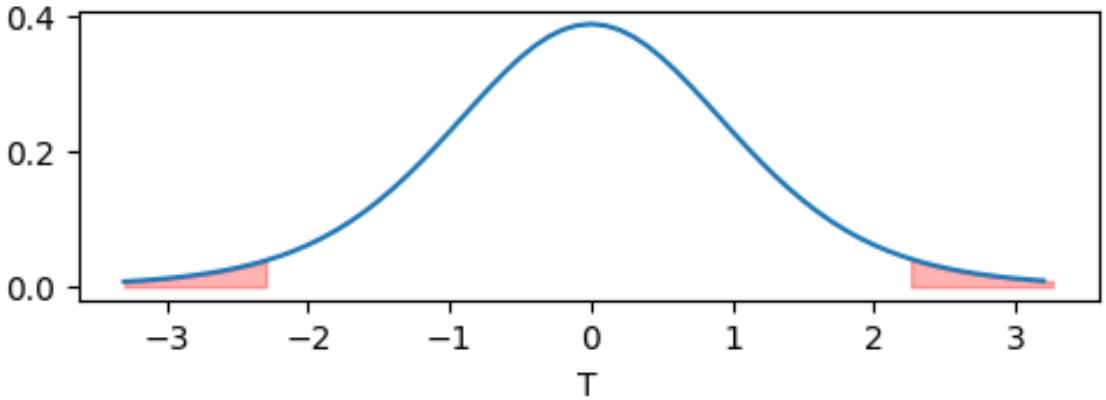
The question is whether the observed value $t_{\text{obs}}$ is "too extreme"?

# Student's t-test, or simply t-test

## Hypothesis test: t-test

$$t_{\text{obs}} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

To answer whether the observed value $t_{\text{obs}}$ is too "extreme," we **test** if $t_{\text{obs}}$ lies within the interval $\pm t_{1-\alpha/2}$ (which depends on our chosen significance level). The test result is either Yes/No, along with the chosen significance level.



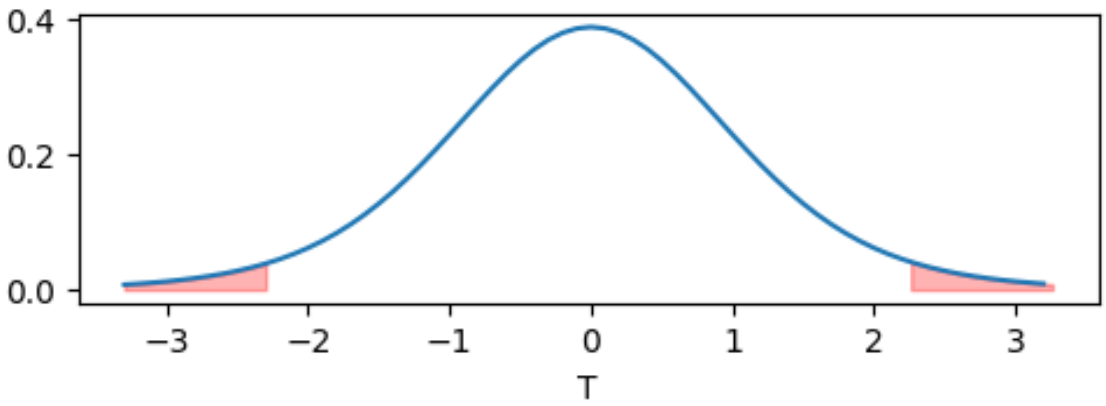

If $|t_{\text{obs}}| > t_{1-\alpha/2}$, we conclude that $|t_{\text{obs}}|$ is too large to believe in the null hypothesis — and we say that we **reject the null hypothesis**.

# Critical values for $t_{obs}$

**Definition 3.31 The critical values**

The $(1-\alpha)100\%$ critical values for the one-sample t-test are the $\alpha/2$- and $1-\alpha/2$-quantiles of the $t$-distribution with $n-1$ degrees of freedom

$$t_{\alpha/2} \text{ and } t_{1-\alpha/2}. \tag{3-23}$$




In the plot above:

What are the critical values $t_{\alpha/2}$ and $t_{1-\alpha/2}$? _____

What do the red areas visualize? _____

# Example – Light Rail

Hypothesis of voltage drop $= 0$:

$$H_0: \ \mu = 0 \ \ (= "\mu_0")$$

where $\mu$ is the average voltage drop.

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Calculate the test statistic:

$$t_{\text{obs}} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{1.23 - 0}{1.62/\sqrt{10}} = 2.39$$

# Example – Light Rail

## Hypothesis of voltage drop = 0:

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where $\mu$ is the average voltage drop.

## Calculate the test statistic:

$$t_{\text{obs}} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{1.23 - 0}{1.62/\sqrt{10}} = 2.39$$

## Compare with $t_{1-\alpha/2}$:

$$t_{1-\alpha/2} = 2.26 \ \ (\alpha = 0.05)$$

## Conclusion:

$|t_{\text{obs}}| > t_{1-\alpha/2}$, and therefore we reject the null hypothesis (at the significance level of 0.05).

# t-test based on critical values

**Method 3.32 The one-sample hypothesis test by the critical value**

A null hypothesis is *rejected* if the observed test-statistic is more extreme than the critical values

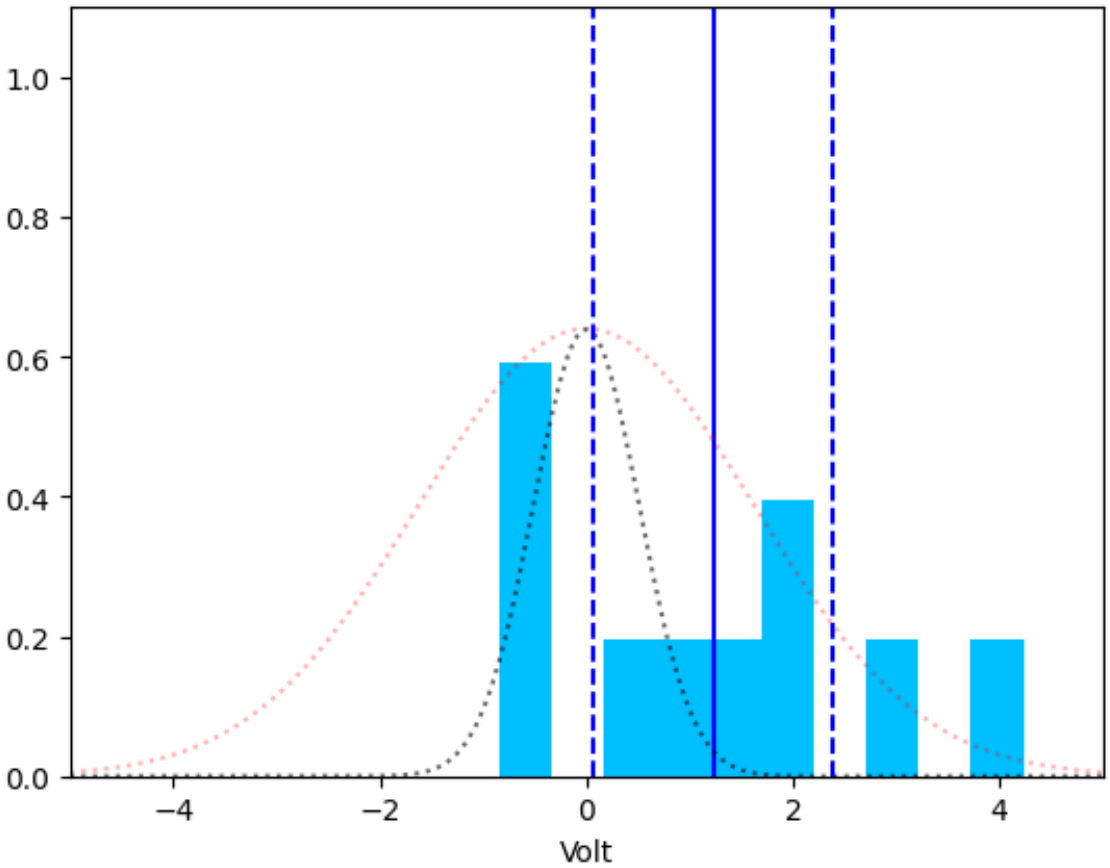
$$\text{If } |t_{\text{obs}}| > t_{1-\alpha/2} \text{ then } \textit{reject}, \tag{3-24}$$

otherwise *accept*.

Note:

$|t_{obs}| > t_{1-\alpha/2}$, is the *same* as saying that $t_{obs}$ lies outside the interval $[t_{\alpha/2}; t_{1-\alpha/2}]$.

# Example - Light Rail




In this case, we reject the null hypothesis.

Note also that $\mu_0$ is just outside the confidence interval (the 95% confidence interval for $\mu$ was $[0.07; 2.9]$).

Kahoot!
(x2)

# Example - light rail

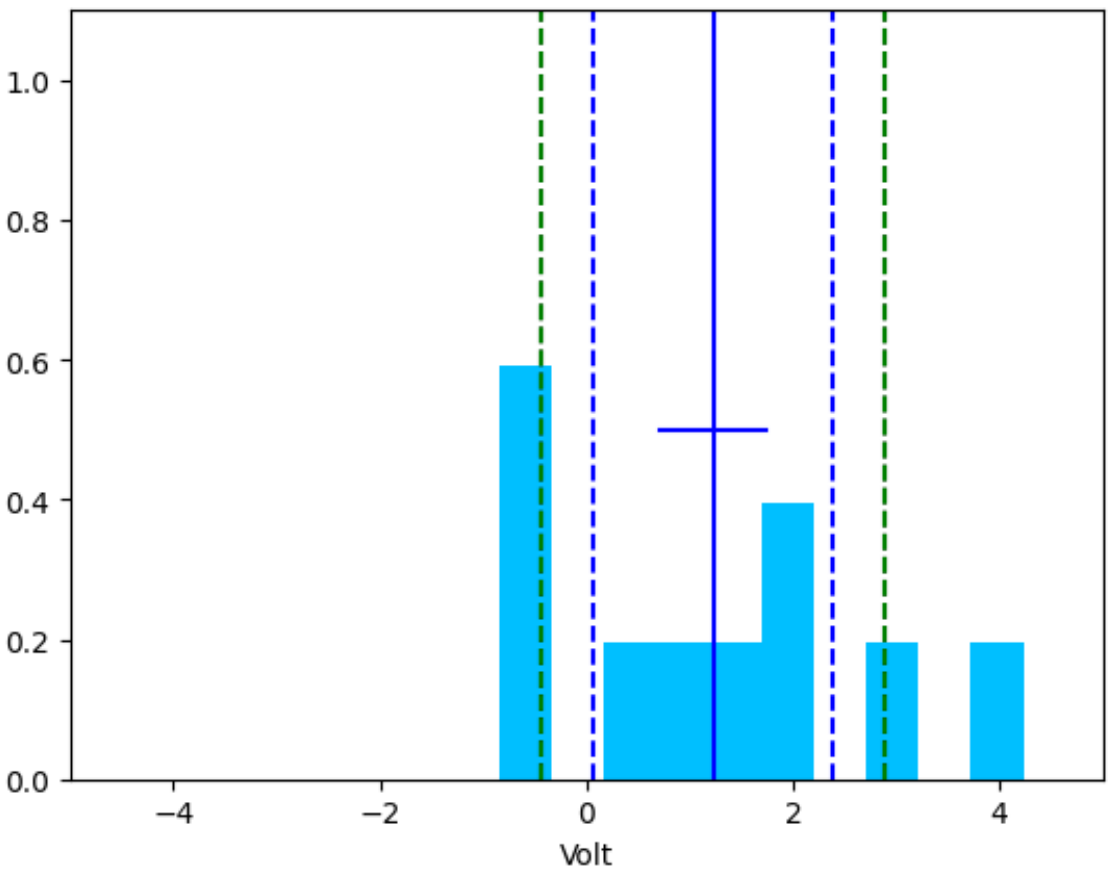
In our example with the light rail we have $t_{obs} = 2.39$.

But the conclusion of the t-test depends on the choice of significance level ($\alpha$):

For $\alpha = 0.05$:
Critical values for $t_{obs}$: $[-2.26, 2.26]$
95% confidence interval for $\mu$: $[0.07; 2.4]$
Conclusion of the t-test: the null hypothesis *is rejected*

For $\alpha = 0.01$:
Critical values for $t_{obs}$: $[-3.25, 3.25]$
99% confidence interval for $\mu$: $[-0.43; 2.9]$
Conclusion of the t-test: the null hypothesis *is not rejected*

# Example - Light Rail



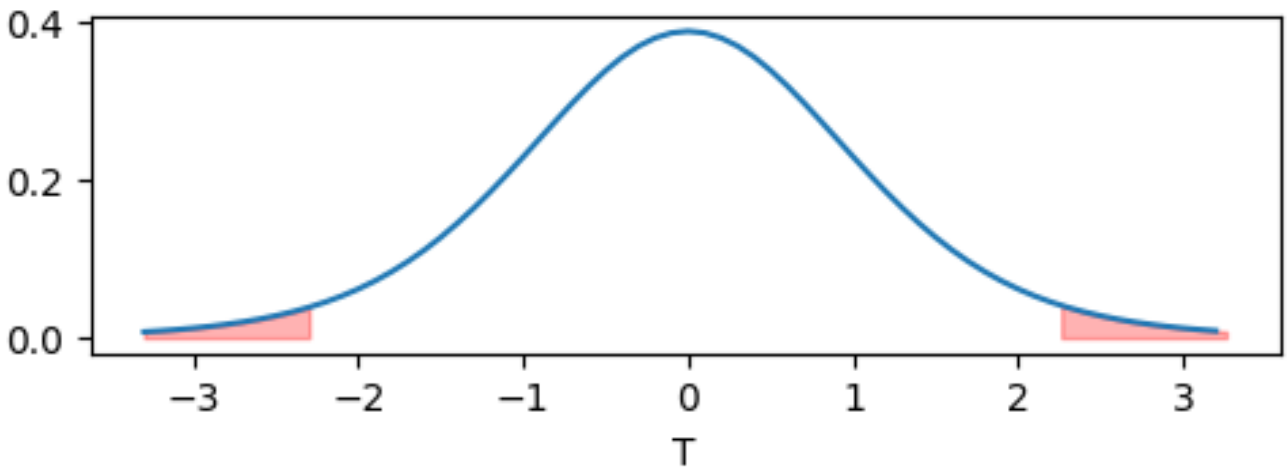

We could also ask - what **significance level** should we have chosen if the value $\mu = 0$ were to be inside the confidence interval?
95% (blue), 99% (green) ... ?

# p-value: The probability of observing $T = t_{obs}$

The next step in the t-test is calculating the p-value:

What significance level should we choose for $t_{\text{obs}}$ to fall within our range of accepted $T$-values?

$$t_{\text{obs}} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$




We now choose the boundaries precisely at $T = \pm t_{\text{obs}}$

That way, $t_{\text{obs}}$ just barely fits within the interval, while values of $T$ that are "more extreme" than $t_{\text{obs}}$ fall outside the interval.

# p-value for t-test with one (quantitative) sample

**Method 3.23 The one-sample $t$-test statistic and the $p$-value**

For a (quantitative) one sample situation, the $p$-value is given by

$$p\text{-value} = 2 \cdot P(T > |t_{\text{obs}}|), \tag{3-20}$$

where $T$ follows a $t$-distribution with $(n-1)$ degrees of freedom.
The observed value of the test statistics to be computed is

$$t_{\text{obs}} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, \tag{3-21}$$

where $\mu_0$ is the value of $\mu$ under the null hypothesis

$$H_0 : \ \mu = \mu_0. \tag{3-22}$$

# p-value compared with $\alpha$

The p-value should be compared with the desired significance level:

The desired significance level is expressed by the value $\alpha$ (e.g. $\alpha = 0.05$).

If $p < \alpha$ we *reject* the null hypothesis.

If $p > \alpha$ we *accept* the null hypothesis.

# p-value

## Definition of p-value:

For a (quantitative) situation with **one sample**, the ***p-value*** is given by:

$$\begin{aligned} p\text{-value} &= 2 \cdot P(T > |t_{\text{obs}}|) \\ &= P(T < -|t_{\text{obs}}|) + P(T > |t_{\text{obs}}|) \end{aligned}$$

where $T$ follows a $t$-distribution with $(n-1)$ degrees of freedom.

## The p-value is compared with the chosen significance level:

The desired significance level is expressed by $\alpha$
(e.g., $\alpha = 0.05$).

We now check if the p-value is less than $\alpha$.

# Example – Light Rail

Hypothesis of no voltage drop:

$$H_0: \ \mu = 0 \ \ (= "\mu_0")$$

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## Hypothesis of no voltage drop:

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## Calculate the test statistic:

$$t_{\text{obs}} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{1.23 - 0}{1.62/\sqrt{10}} = 2.39$$

# Example – Light Rail

## Hypothesis of no voltage drop:

$$H_0: \ \mu = 0 \ \ (= "\mu_0")$$

where $\mu$ is the average voltage drop.

## Calculate the test statistic:

$$t_{\text{obs}} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{1.23 - 0}{1.62/\sqrt{10}} = 2.39$$

## Calculate the $p$-value:

$$2P(T > 2.39) = 0.0404$$

## Conclusion:

If we choose $\alpha = 0.05$, we get $p < \alpha \rightarrow$ We reject the null hypothesis.

However, if we had chosen $\alpha = 0.04$, we would have accepted the null hypothesis.

# Definition and Interpretation of p-value (General)

The $p$-value expresses *evidence* against the null hypothesis – Table 3.1:

| | |
|---|---|
| $p < 0.001$ | Very strong evidence against $H_0$ |
| $0.001 \leq p < 0.01$ | Strong evidence against $H_0$ |
| $0.01 \leq p < 0.05$ | Some evidence against $H_0$ |
| $0.05 \leq p < 0.1$ | Weak evidence against $H_0$ |
| $p \geq 0.1$ | Very weak or no evidence against $H_0$ |

Definition 3.22 of $p$-value:

The $p$-value is the probability of observing a test statistic that is **at least as extreme** as the observed test statistic. This probability is calculated under the assumption that the null hypothesis is true.

# Python

- Go to today's Python notebook in VS Code
  - "Example: Voltage drop"



# Statistical Significance and Terminology

We say that we are *performing a hypothesis test* when we decide to reject or accept a null hypothesis based on data.

A null hypothesis is *rejected* if $p$-value $< \alpha$ ($\alpha$ is chosen in advance).

Otherwise, the null hypothesis is said to be *'accepted'*. It is more accurate (and preferable) to say that the null hypothesis cannot be rejected.

*Statistical Significance*: An *effect* is said to be *(statistically) significant* if the $p$-value is less than the significance level $\alpha$.
This terminology is most meaningful when the null hypothesis is: $H_0: \ \mu = \mu_0 = 0$, which translates to the null hypothesis of "no effect."

It can also be said that $\mu$ is *significantly different from* $\mu_0$.

Sometimes, we also say that we 'accept' the *alternative hypothesis*: $\mathrm{H}_A : \mu \neq \mu_0$

# Steps in a t-test – An Overview

In general, a t-test consists of the following steps:

1. Formulate the null hypothesis (and alternative hypothesis) and choose a significance level $\alpha$ (set the "risk level").
2. Calculate the test statistic value from the observed data.
3. Compute the p-value based on the test statistic compared to the appropriate distribution.
4. Compare the p-value with the significance level $\alpha$ and conclude.

# t-test in Python

Calculate $t_{obs}$ "by hand" and find the p-value in the correct $t$ distribution:

- `2*stats.t.cdf(-tobs, df=n-1))`

You can also use the function:

- `stats.ttest_1samp(...)` (from the `Scipy.stats` package)

The same function can also compute a confidence interval directly:

- `stats.ttest_1samp(...).confidence_interval(...)`

- Go to the Python notebook "voltage_drop.ipynb" in VS Code



# One-sided vs Two-sided Tests

So far, it has been implied that the test is two-sided (non-directional):

The alternative to $H_0: \mu = \mu_0$ is $H_A: \mu \neq \mu_0$.

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There may be other situations, such as one-sided (= directional) alternative hypotheses:

The alternative to $H_0: \ \mu = \mu_0$ is $H_A: \ \mu < \mu_0$.

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There may be other situations, such as one-sided (= directional) alternative hypotheses:

The alternative to $H_0: \ \mu = \mu_0$ is $H_A: \ \mu < \mu_0$.

In this course, we stick to the two-sided test (non-directional)!

# Example with t-test

A t-test is often used to test if there is a **difference** between two things.

Example:
In a study, we want to examine whether there is a significant difference in calorie intake throughout women's menstrual cycle.

We measure the **difference in calorie intake** in 11 women before and after menstruation.

The data (difference in calories) is as follows: 1350, 1250, 1755, 1020, 745, 1835, 1540, 1540, 725, 1330, 1435

What is a relevant null hypothesis?

Do we believe there is a significant difference?

How should we conduct our statistical analysis of this data?




(x4)

# Python:

- Go to today's Python notebook in VS Code
  - "Example: Difference in calorie intake"



# Overview


1. Summary from last week
2. Model Control
   - Q-Q plot
   - Transformation towards Normality
3. Hypothesis Testing
   - The Null Hypothesis and the Alternative Hypothesis
   - t-test
   - p-value
   - One-sided vs Two-sided Tests
4. Type I and Type II Errors
   - Requirements for precision
   - Power and Sample Size - One Sample

# Type I and Type II Errors

## There are two types of errors (only one at a time)

Type I: Rejecting $H_0$ when $H_0$ is true.
Type II: Accepting (not rejecting) $H_0$ when $H_1$ is true.

Type I error is called a false positive.
Type II error is called a false negative.
In this terminology, $H_0 =$ "negative" ("no effect")
and $H_1 =$ "positive" (there is an "effect")

## The risk of the two types of errors is usually referred to as:

$$P(\text{Type I error}) = \alpha$$

$$P(\text{Type II error}) = \beta$$

# Courtroom Analogy

A person is brought before a court:

A person is being prosecuted under a specific charge.
The null and alternative hypotheses are:

$H_0$ : The person is innocent.

$H_1$ : The person is guilty.

# Courtroom Analogy

**A person is brought before a court:**

A person is being prosecuted under a specific charge.
The null and alternative hypotheses are:

$H_0$ : The person is innocent.

$H_1$ : The person is guilty.

**Not being proven guilty is not the same as being proven innocent:**

In other words:
*Accepting* a null hypothesis is not a statistical proof that the null hypothesis is true!

# Errors in Hypothesis Testing

**Theorem 3.39: The significance level is the risk of making a Type I error**

The significance level $\alpha$ in hypothesis tests is the risk of a Type I error:

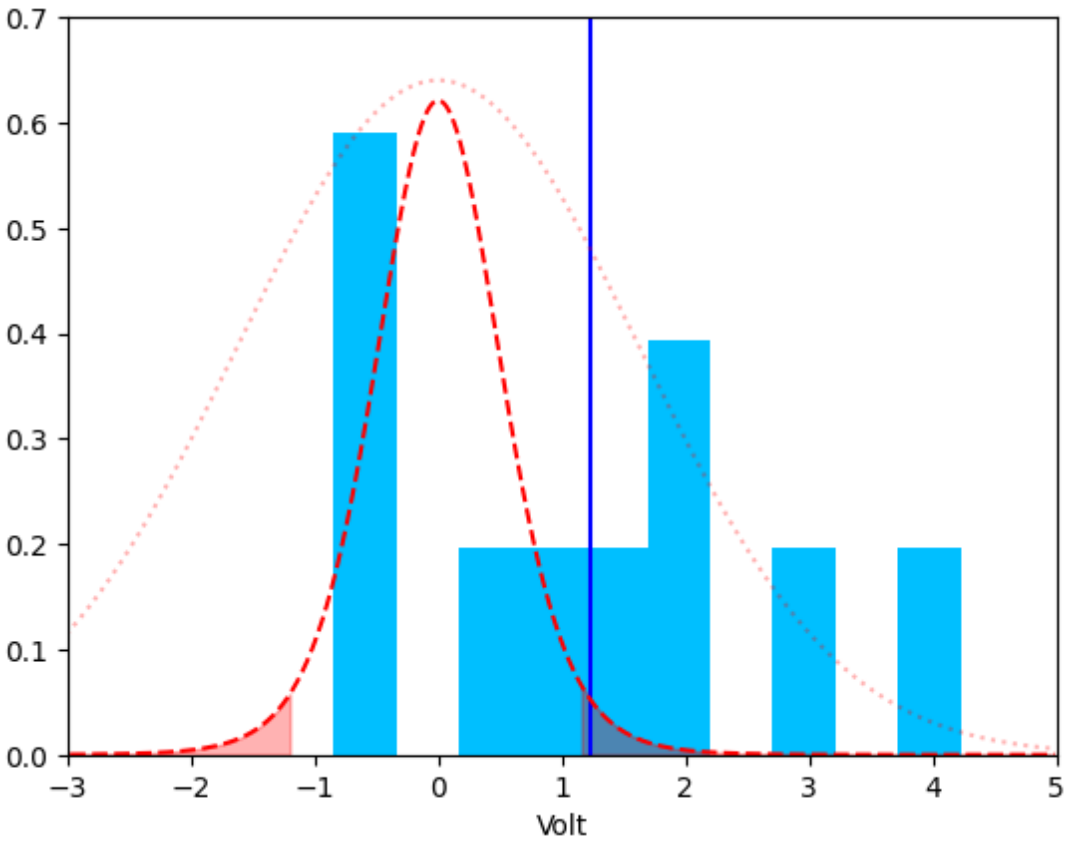
$$P(\text{Type I error}) = P(\text{Rejecting } H_0 \text{ when } H_0 \text{ is true}) = \alpha$$

# Errors in Hypothesis Testing

**Theorem 3.39: The significance level is the risk of making a Type I error**

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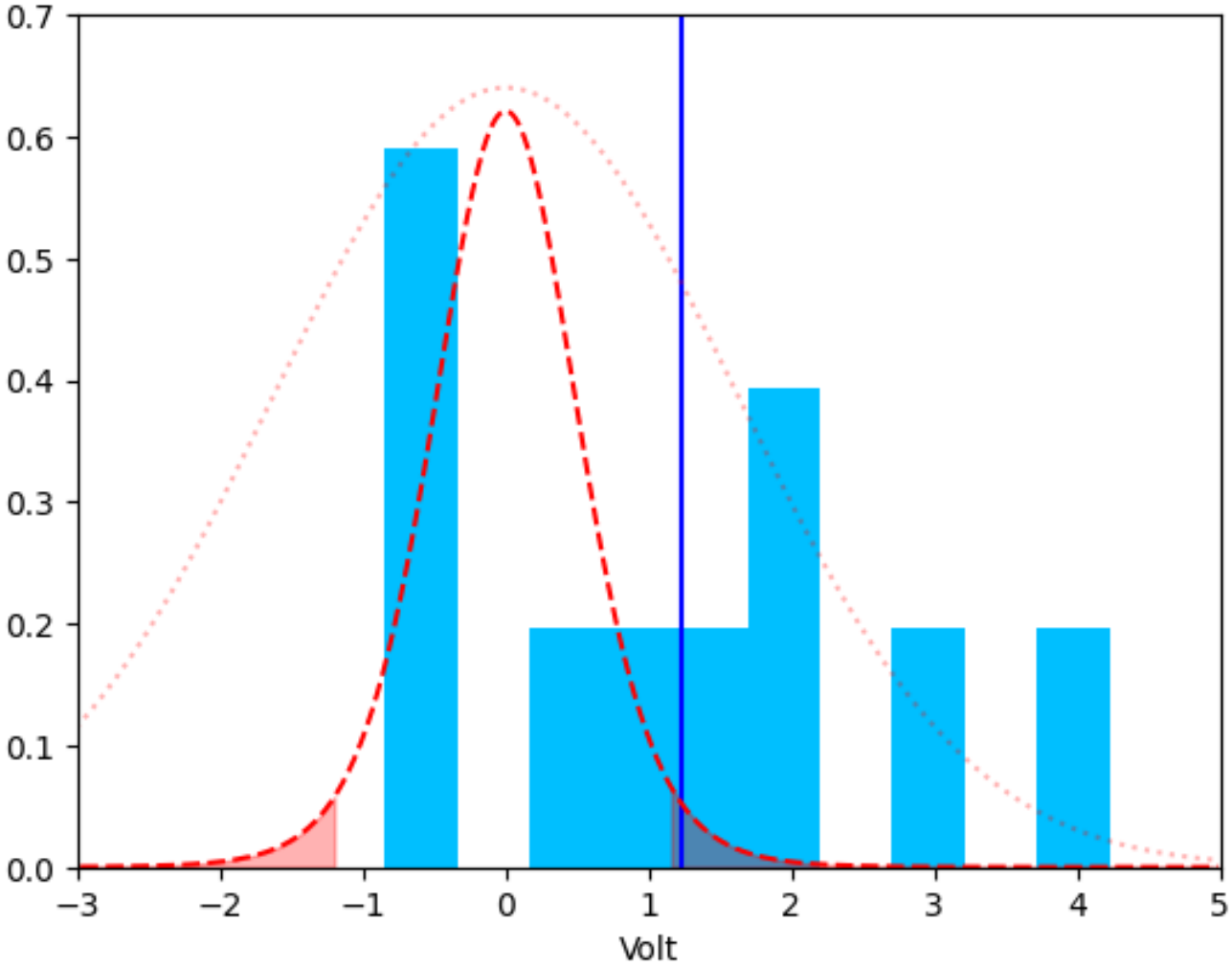


# Errors in Hypothesis Testing

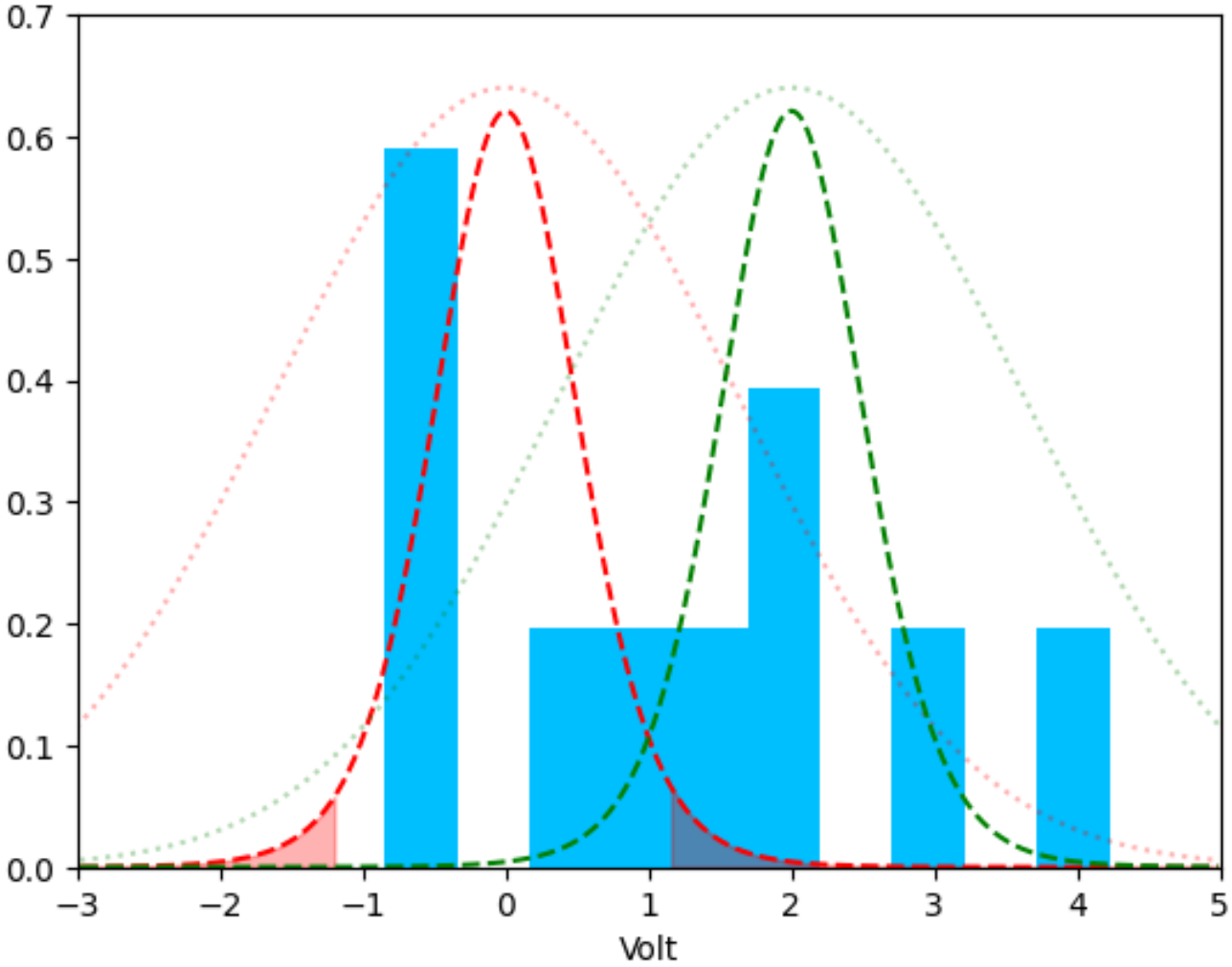
Two possible truths against two possible conclusions:

| | Rejecting $H_0$ | Not rejecting $H_0$ |
|---|---|---|
| $H_0$ is true | Type I error ($\alpha$) | Correct acceptance of $H_0$ |
| $H_0$ is false | Correct rejection of $H_0$ | Type II error ($\beta$) |

# Type I Error



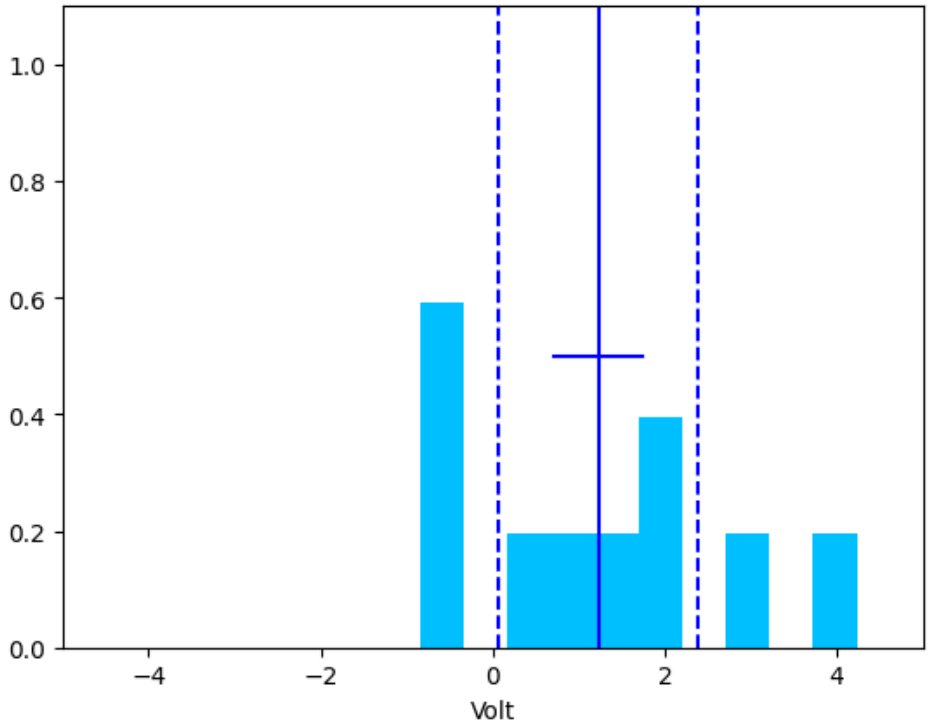
# Type I and Type II Errors




Smaller $\alpha$ = Larger $\beta$ (and vice versa)

# Example - light rail

Let us look at data from the light rail example again. The data measures the voltage drop across a point where the voltage drop should be zero on average.




From the sample (n=10):

$\bar{x} = 1.23$

$s = 1.62$

confidence interval:

$\bar{x} \pm t_{1-\alpha/2} \cdot (s/\sqrt{n}) =$

$1.23 \pm 1.16$

$[0.07; 2.9]$

In this case we have: $\hat{\mu} = 1.23 \pm 1.16$ volt $(\alpha = 0.05)$.

We say that the **"Margin of Error" (ME)** is 1.16 (volt).

# Example - light rail

The conclusion from the sample was that we reject the null hypothesis – thus we conclude that the voltage drop is NOT zero (on average).

But the boss is not satisfied and says that we must investigate the matter further.
Our margin of error of $\pm 1.16$ volts is too large.

We therefore plan to conduct a new sample.

How can we do it better than the first time?

What can we do to improve our estimate of $\mu$?

Kahoot!
(x1)

# Experimental design with requirements for precision

*Margin of error* (ME) is defined as

$$ME = t_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}$$

For sufficiently large samples, we can approximate $t_{1-\alpha/2}$ (from the t-distribution) with $z_{1-\alpha/2}$ (from the normal distribution).

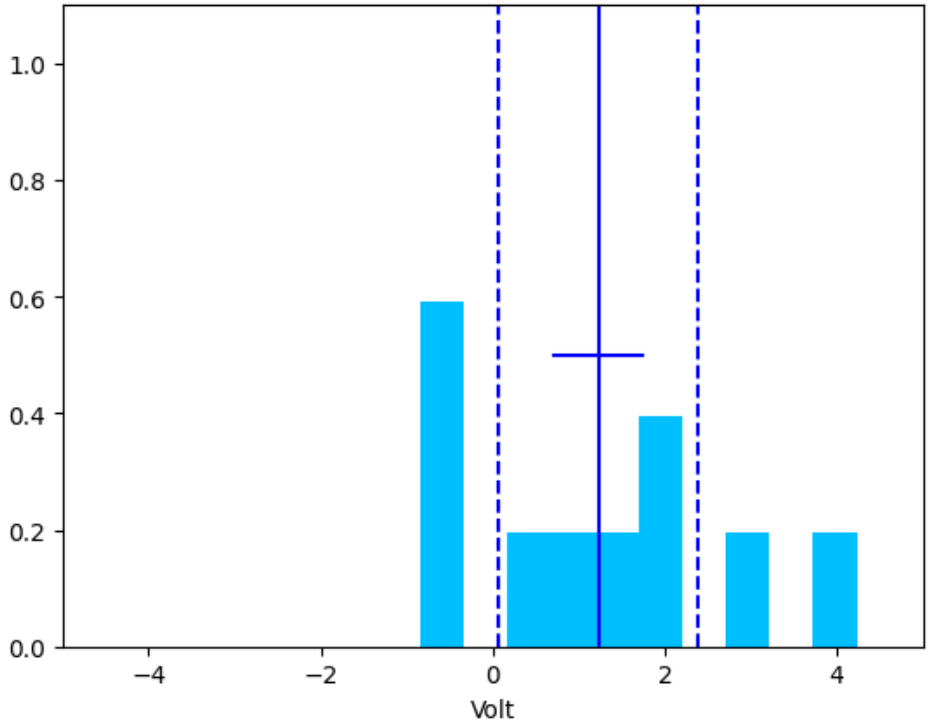
## Method 3.63: Sample size for the confidence interval based on a sample

If $\sigma$ is known, or estimated to be a certain value, then we can calculate the sample size required to achieve a given margin of error, with probability $1-\alpha$.

$$n = \left(\frac{z_{1-\alpha/2}\cdot\sigma}{ME}\right)^2$$

# Example - light rail

Let us estimate the sample size if we want ME at most 0.5 volts.
We assume that $s = 1.62$ is a reasonable guess for the true $\sigma$.




$$n = \left(\frac{z_{1-\alpha/2} \cdot \sigma}{ME}\right)^2$$
$$= \left(\frac{1.96 \cdot 1.62}{0.5}\right)^2$$
$$= 40.3$$

Since we cannot make 40.3 measurements, we round up to 41 (in the new sample $n = 41$).

# Planning (Power and Sample Size)

We will now talk about experimental planning and "power".

For this, we need to consider the two types of errors that I introduced earlier

Type I: Rejection of $H_0$ when $H_0$ is true.

$$P(\text{Type I error}) = \alpha$$

Type II: Acceptance (non-rejection) of $H_0$ when $H_1$ is true.

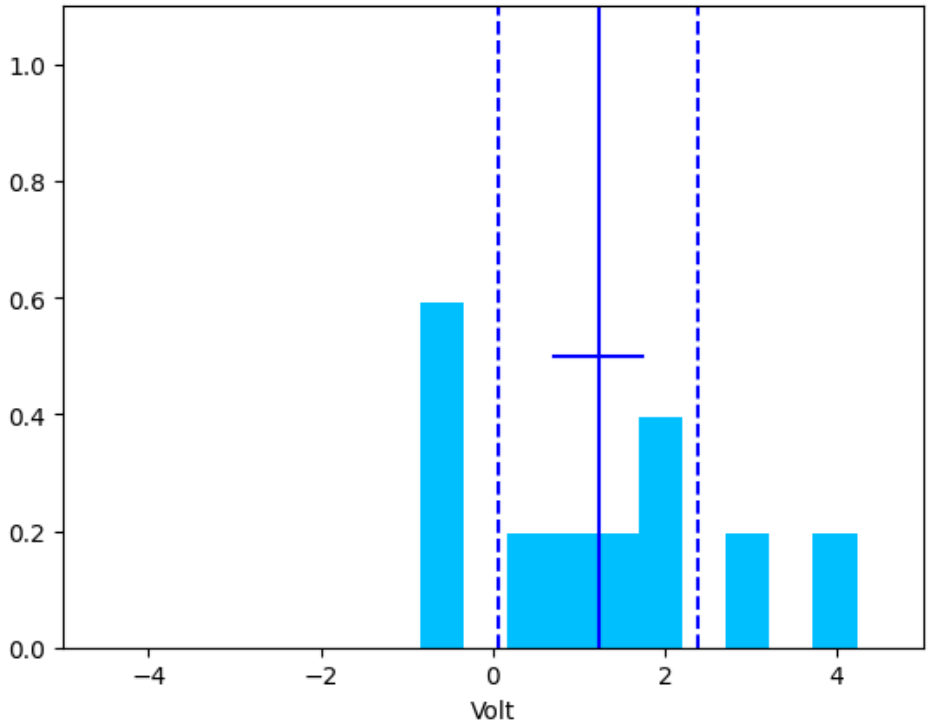
$$P(\text{Type II error}) = \beta$$

# Example: light rail

The boss is now becoming a bit more demanding. She says that if the true average voltage drop is $\pm 0.5$ volts (or more), then we would like to know that.

How should we proceed?

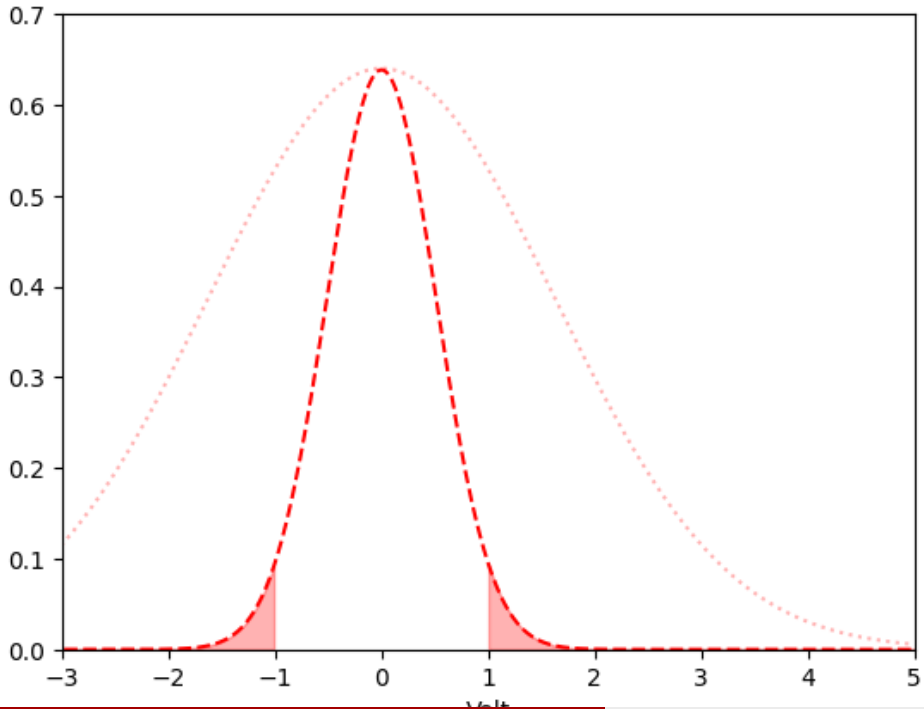
We will now go through some different **scenarios**.



# Example: light rail

**Scenario 1: The null hypothesis is true**

- *If* the true average voltage drop *is* zero
- and we take a new set of measurements – this time with larger n
- then the risk of rejecting the null hypothesis by mistake will be $\alpha$ (e.g. 0.05, or 5%)




The values of $\bar{x}$ that we potentially accept as being "not significantly different from zero" lie in the interval $0 \pm ME$. The width of this interval depends on the sample size ($n$).

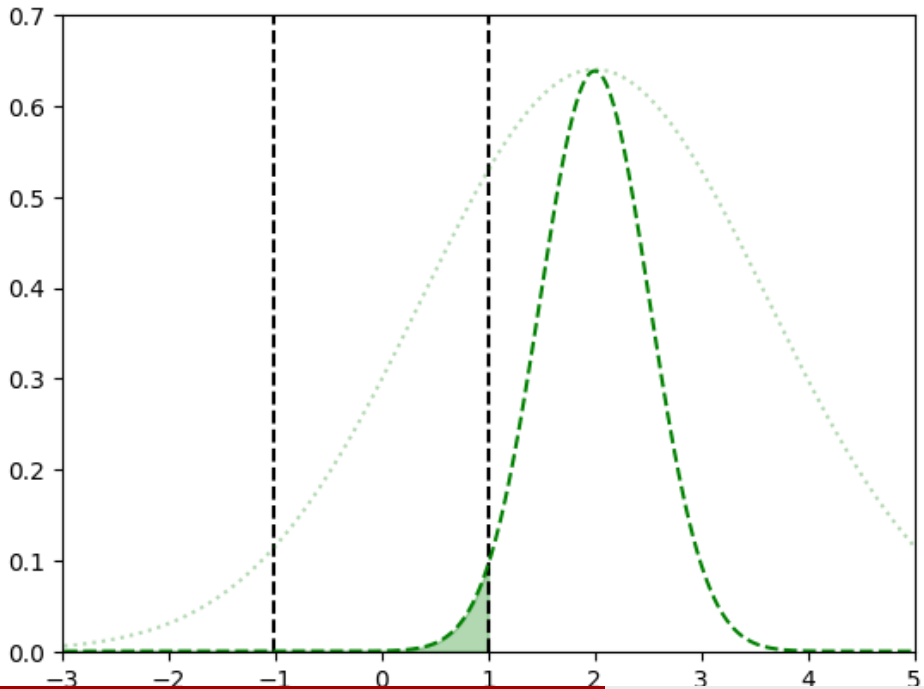
Kahoot!

(x2)

# Example: light rail

**Scenario 2: The alternative hypothesis is true**

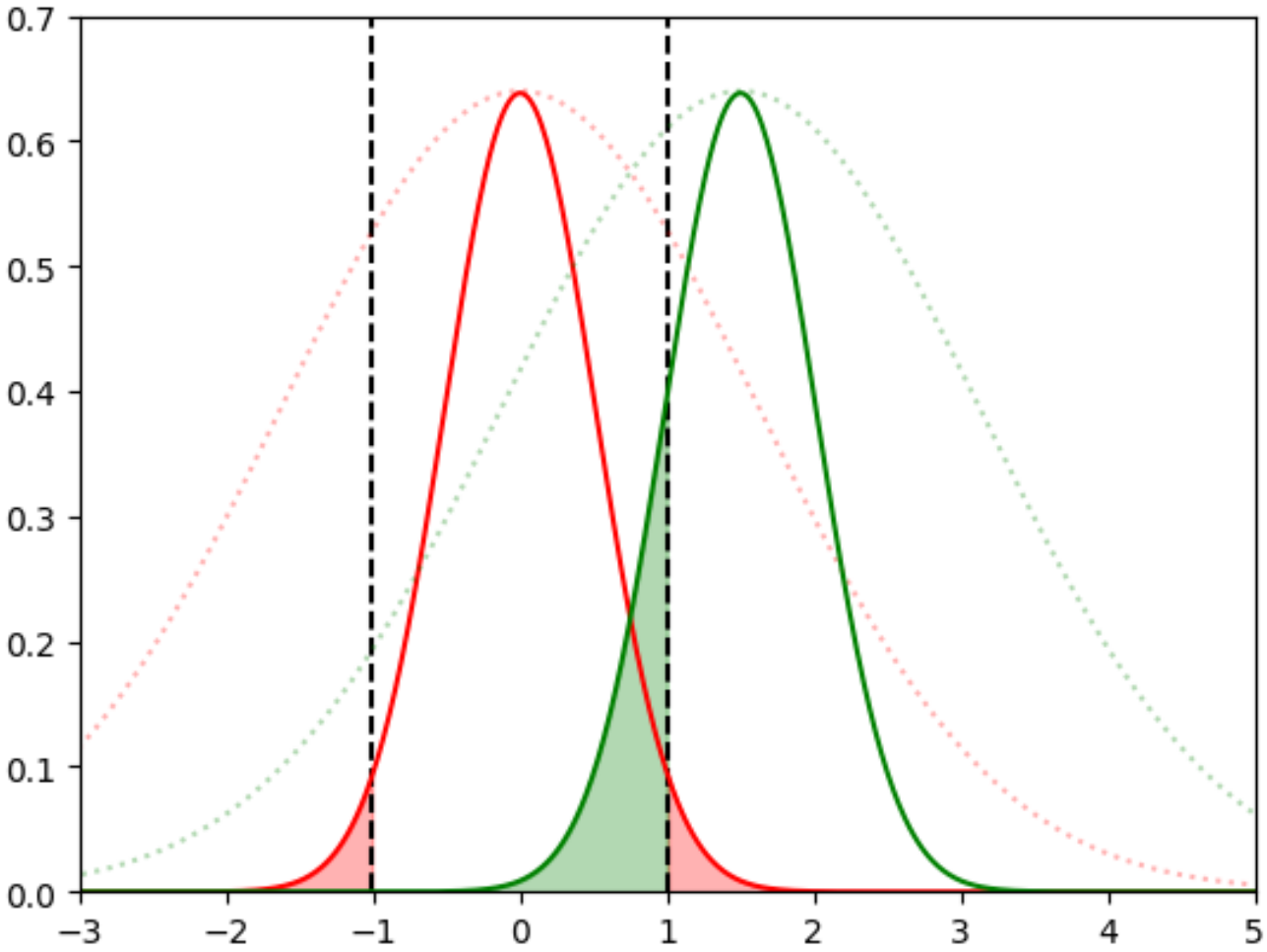
- *If* the true average voltage drop is actually something other than zero – the true value is $\mu_1$
- and we take a new round of measurements – this time with larger n
- What is then the risk (probability $\beta$) of accepting (not rejecting) the null hypothesis?




Yes, it depends on...
...how large $\mu_1$ actually is

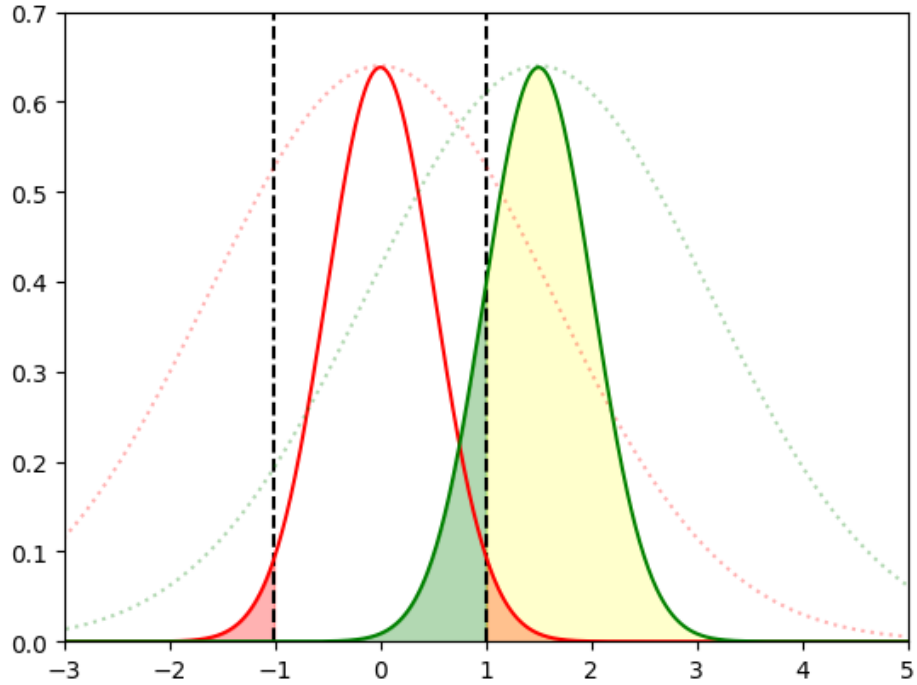
Kahoot!
(x2)

# Type I and Type II errors




$\alpha$: the red area $\beta$: the green area $(1-\beta)$ is called **power**

# Experimental Planning: Power




- **Power** $= 1 - \beta$
- $P(\text{Type II error}) = \beta$
- Power = "The probability of a correct rejection of $H_0$".
- Power = "The probability of detecting an (alleged) effect".
- Typical values for desired power are 80%, 90%, .. (something relatively large)
- In practice: Use a scenario-based approach

Kahoot! (x3)

# Experimental Design: Power

If we know (or assume) four out of the following five quantities, then we can find the missing one:

- Sample size, $n$.
- Significance level, $\alpha$, at which we test.
- Difference in mean (effect size), $\mu_0 - \mu_1$.
- Population standard deviation, $\sigma$.
- Power, $1 - \beta$.

(blackboard)

# Experimental Planning: Sample Size $n$

## The Big Question: How large should $n$ be?

We need enough observations to be able to detect a relevant effect with high power $1-\beta$ (typically at least 80%):

# Experimental Planning: Sample Size $n$

### The Big Question: How large should $n$ be?

We need enough observations to be able to detect a relevant effect with high power $1-\beta$ (typically at least 80%):

### Method 3.65: Formula for Sample Size with One Sample

For a one-sample t-test, where $\alpha$, $\beta$, and $\sigma$ are given:

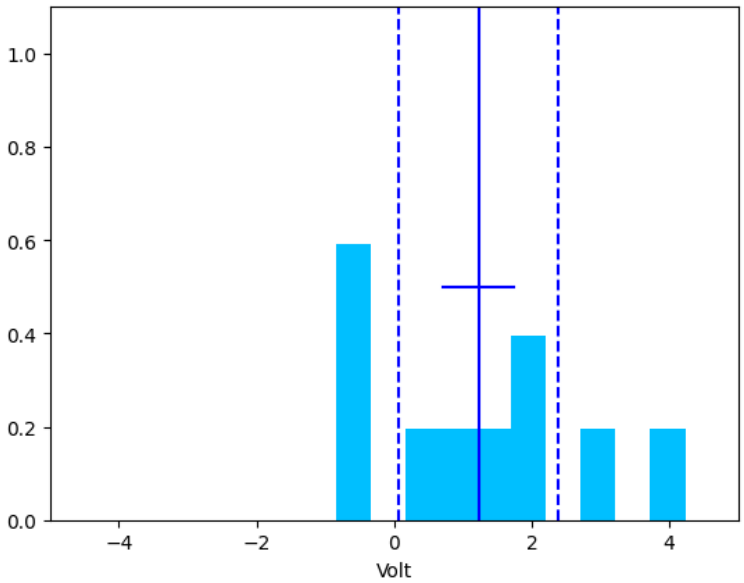
$$n = \left( \sigma \frac{z_{1-\beta} + z_{1-\alpha/2}}{\mu_0 - \mu_1} \right)^2.$$

Here, $\mu_0 - \mu_1$ is the difference in means that we want to measure, while $z_{1-\beta}$ and $z_{1-\alpha/2}$ are quantiles of the standard normal distribution.

# Example: light rail

We now tell the boss that we can design the next sample such that we will have an 80% probability of "detecting" a voltage drop of $0.5$ volts (or more), at a significance level $\alpha = 0.05$.

To calculate the required sample size, we use (as an informed guess): $\sigma = 1.62$, corresponding to the variation we observed in our previous measurements.




We then arrive at the conclusion that the next sample must have 83 measurements:

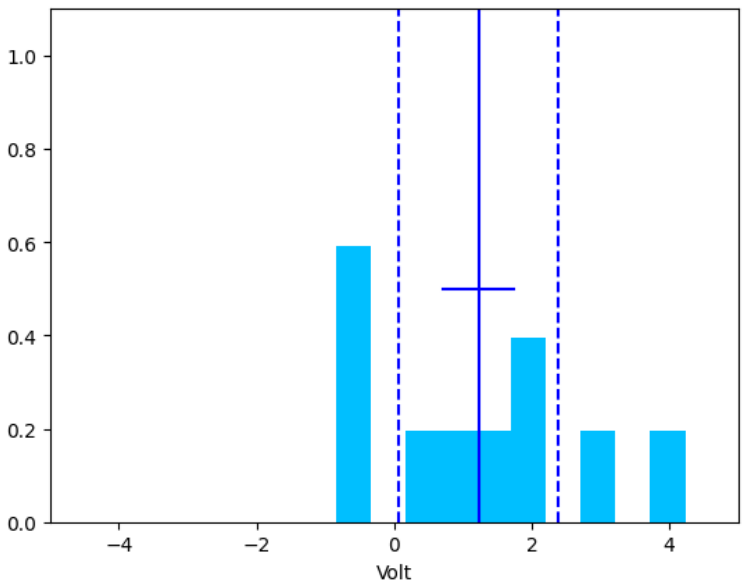
$$
\begin{aligned}
n &= \left( \sigma \frac{z_{1-\beta} + z_{1-\alpha/2}}{\mu_0 - \mu_1} \right)^2 \\
&= \left( 1.62 \frac{z_{0.80} + z_{0.975}}{0.5} \right)^2 \\
&= 82.4
\end{aligned}
$$

# Example: light rail

"83 measurements are too expensive," says the boss.

"Can we make do with 60 measurements?"

Yes... – but then our experiment will have less power.




New power calculation:

$$n = \left( \sigma \frac{z_{1-\beta} + z_{1-\alpha/2}}{\mu_0 - \mu_1} \right)^2$$

$$\Rightarrow$$

$$z_{1-\beta} = \sqrt{60 \frac{0.5^2}{1.62^2}} - z_{0.975}$$

$$= 0.43$$

corresponding to a power of 0.67

# Python

- Go to today's Python notebook in VS Code
  - "Example: Power calculations"



# Agenda

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